

DATA ANALYTICS PORTFOLIO PROJECT

Superstore Sales Analysis & Customer Churn Analysis

Executive Sales Dashboard & Business Insights Report

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End-to-end analytics pipeline:

Python (Pandas) for data cleaning & EDA → Microsoft SQL Server for structured query analysis → Power BI for executive dashboarding

Project Overview

This project analyzes four years (2014–2017) of order-level transactional data from a national retail superstore to answer six core business questions: total sales and profit, regional performance, top-revenue products, top customers, and how sales trend over time. The project was built as a complete analytics pipeline — Python for cleaning and exploratory analysis, Microsoft SQL Server for structured querying, and Power BI for executive dashboarding — mirroring the toolchain used in real business intelligence roles.

Business Questions

- What are the total sales and total profit?
- Which regions perform best, and where is margin weakest?
- Which products and sub-categories generate the most revenue — and the most loss?
- Who are the top customers by spend?
- How do sales change over time, month to month and year over year?
- How does discounting affect profitability?

\$2.30M Total Sales	\$286K Total Profit	5,009 Total Orders
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Overall Profit Margin: 12.5% | Average Order Value: \$459 | Units Sold: 37,873

Phase 1–2 — Data Cleaning & Exploratory Analysis (Python)

Objective: Load and validate the raw order data, engineer time-based features for trend analysis, and produce the core descriptive statistics that anchor every later phase of the project.

Method

- **Load & inspect:** Loaded the raw dataset with Pandas and inspected structure, data types, and value ranges across all 24 fields.
- **Data quality check:** Verified zero missing values and zero duplicate rows across all 9,994 order line items — the dataset required no imputation or de-duplication.
- **Feature engineering:** Extracted Year, Month, and Quarter from Order Date to support time-series and seasonal analysis in both SQL and Power BI.
- **Core analysis:** Computed total sales, total profit, total orders, and grouped summaries by category, region, customer, and month — exported as cleaned_superstore.csv for downstream SQL and Power BI use.

```
import pandas as pd
df = pd.read_csv('Superstore.csv')
df.isnull().sum().sum()      # 0 missing values
df.duplicated().sum()        # 0 duplicate rows
df['Year'] = pd.to_datetime(df['Order Date']).dt.year
```

```
df['Month'] = pd.to_datetime(df['Order Date']).dt.strftime('%b')
df['Quarter'] = pd.to_datetime(df['Order Date']).dt.quarter
df.to_csv('cleaned_superstore.csv', index=False)
```

Findings

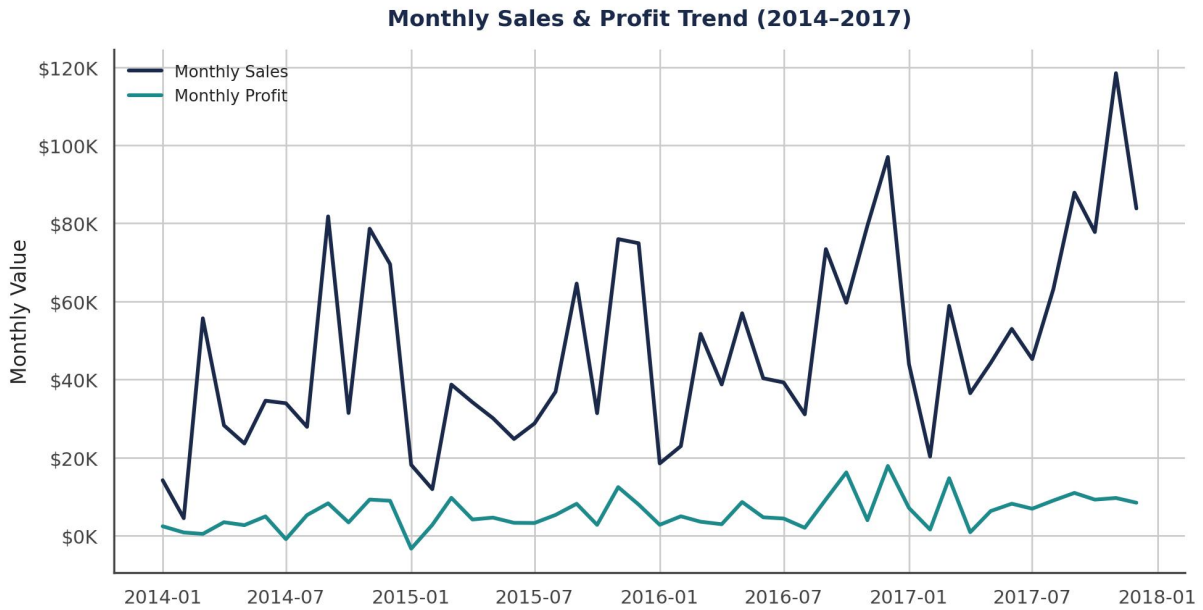


Figure 1. Sales grew steadily from 2014 to 2017 (+51% over the period) with a strong recurring Q4 seasonal spike — profit tracks sales but stays comparatively flat, signaling a margin pressure as volume scales.

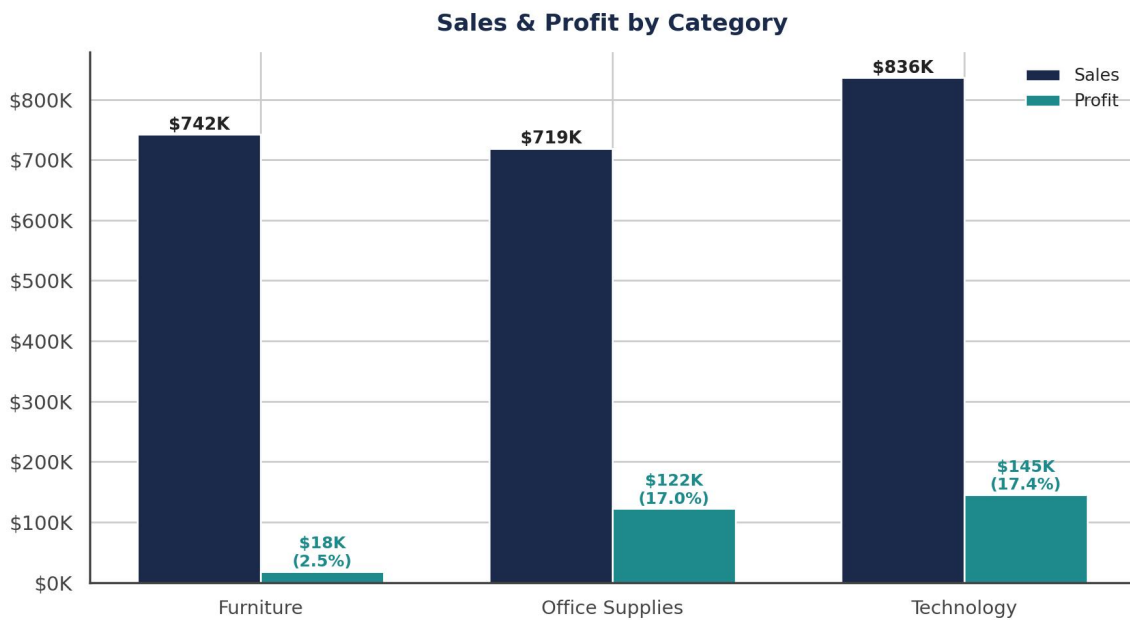


Figure 2. Furniture and Office Supplies generate comparable revenue to Technology, but Furniture converts only 2.5% of sales to profit versus ~17% for the other two categories.

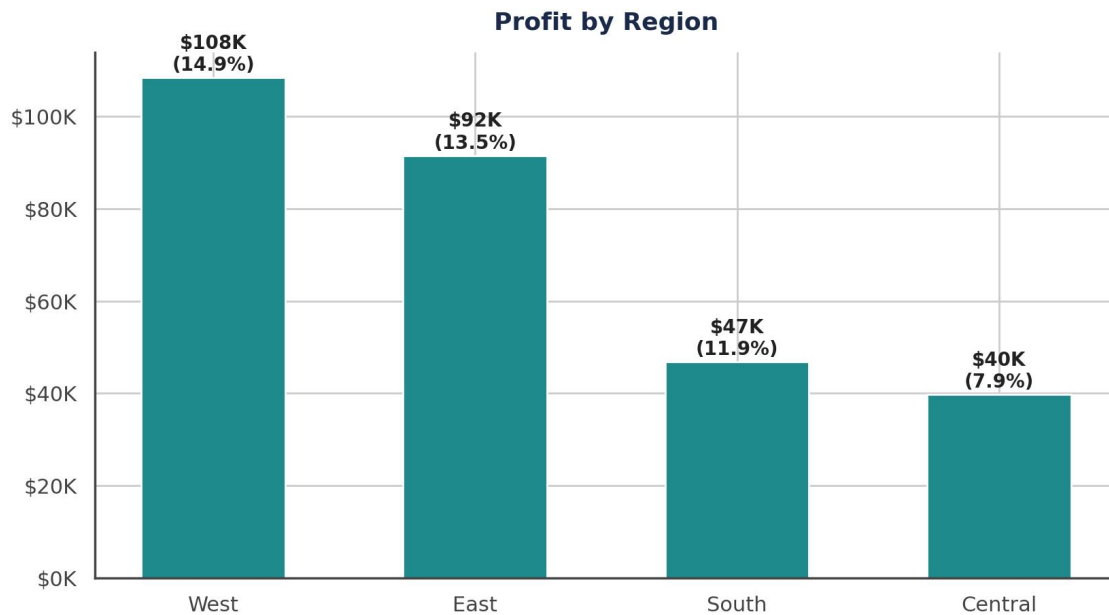


Figure 3. The West region leads on both profit (\$108K) and margin (14.9%); Central trails on margin (7.9%) despite mid-pack sales — a strong candidate for a regional pricing or cost review.

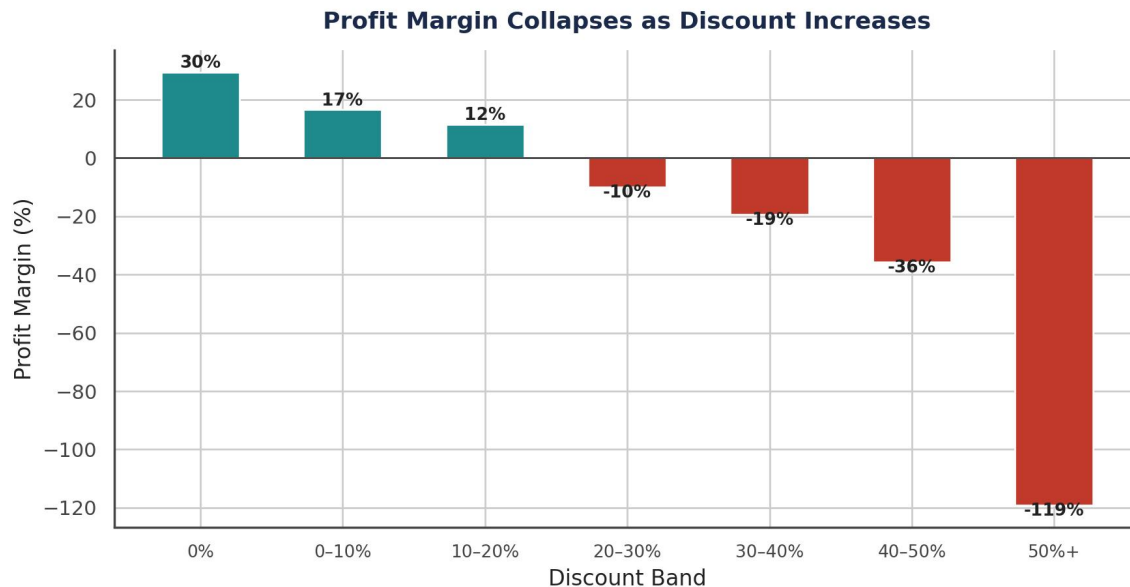


Figure 4. Profit margin falls in a near-straight line as discount increases, turning sharply negative beyond 20% — orders discounted 50%+ lose money at a rate of -119% of sale price.

Key Insights

- **Furniture is a volume category, not a profit category.** Furniture's 2.5% margin versus Technology's 17.4% means a dollar of Furniture revenue is worth roughly 7x less in profit — pricing or cost-of-goods on Furniture deserves review.
- **Discounting above 20% is actively unprofitable.** Every discount band above 20% destroys value — the 50%+ tier loses \$1.19 for every \$1.00 of revenue. Capping discounts at 20% would protect margin on a large share of at-risk orders.

- **Growth is real, but margin discipline matters as it scales.** 2017 sales (\$733K) were 51% higher than 2014 (\$484K)
 - the business is growing, but profit growth needs to be checked against this same baseline to confirm margin isn't eroding as the company scales.

Phase 3 — SQL Server Analysis

Objective: Reproduce and structure the core business metrics in Microsoft SQL Server, demonstrating the ability to move from a flat file to a queryable relational structure suitable for BI tools and ad-hoc analyst queries.

Database & Import

```
CREATE DATABASE SuperstoreDB;  
-- cleaned_superstore.csv imported via SSMS Import Wizard  
-- into dbo.Superstore
```

Core Queries

— Total Sales & Total Profit

```
SELECT SUM(Sales) AS Total_Sales FROM Superstore; -- 2,297,200.86  
SELECT SUM(Profit) AS Total_Profit FROM Superstore; -- 286,397.02
```

— Sales by Category

```
SELECT Category, SUM(Sales) AS Total_Sales, SUM(Profit) AS Total_Profit  
FROM Superstore  
GROUP BY Category  
ORDER BY Total_Sales DESC;
```

— Profit by Region

```
SELECT Region, SUM(Profit) AS Total_Profit,  
       SUM(Profit)/SUM(Sales) AS Profit_Margin  
FROM Superstore  
GROUP BY Region  
ORDER BY Total_Profit DESC;
```

— Top 10 Customers & Top 10 Products

```
SELECT TOP 10 [Customer Name], SUM(Sales) AS Total_Sales  
FROM Superstore GROUP BY [Customer Name] ORDER BY Total_Sales DESC;  
  
SELECT TOP 10 [Product Name], SUM(Sales) AS Total_Sales  
FROM Superstore GROUP BY [Product Name] ORDER BY Total_Sales DESC;
```

— Monthly Sales Trend

```
SELECT YEAR(Order_Date) AS Yr, MONTH(Order_Date) AS Mo, SUM(Sales) AS Total_Sales  
FROM Superstore  
GROUP BY YEAR(Order_Date), MONTH(Order_Date)  
ORDER BY Yr, Mo;
```

Findings

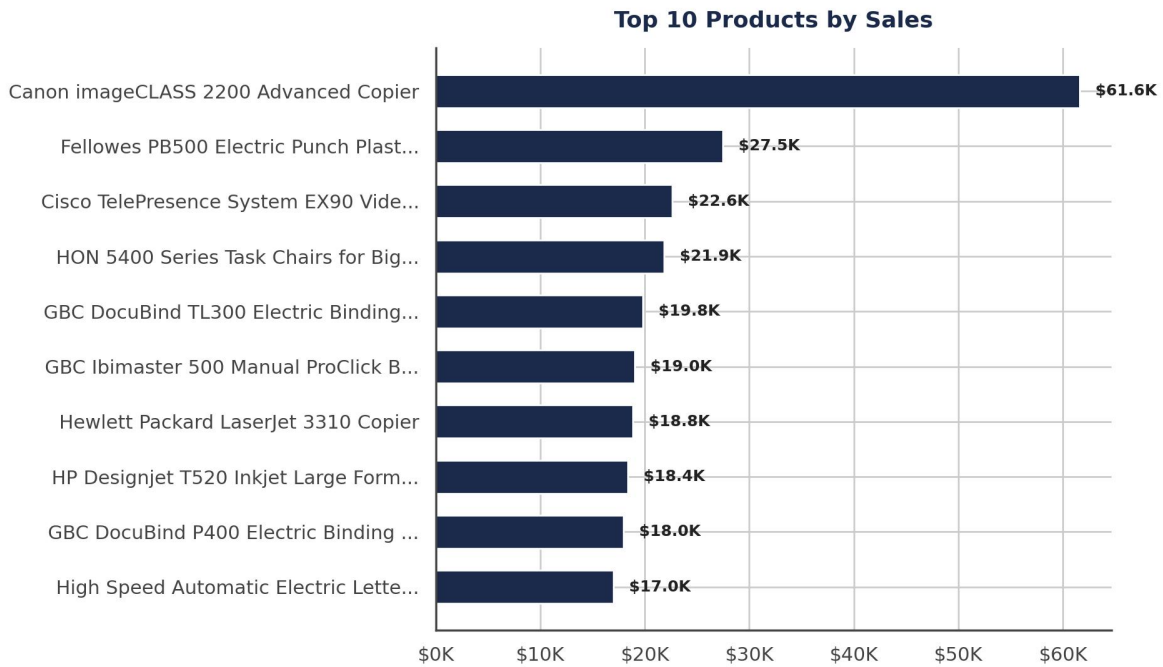


Figure 5. The Canon imageCLASS 2200 Copier is the single highest-revenue product (\$61.6K) — more than double the next product on the list.

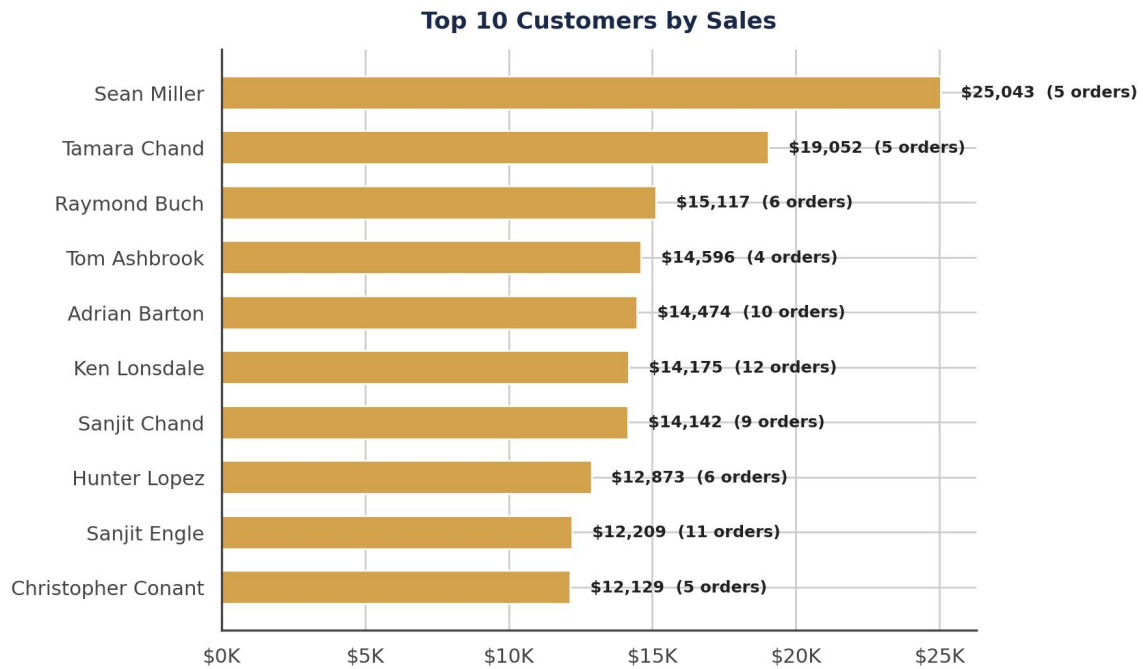


Figure 6. Sean Miller is the top customer by spend (\$25K across 5 orders); several other top-10 customers spread similar value across many more, smaller orders.

Phase 4 — Power BI Dashboard Development

Objective: Translate the SQL-validated metrics into an interactive executive dashboard with KPI cards, trend and comparison visuals, and slicers for self-service filtering by year, region, segment, and category.

DAX Measures

```
Total Sales    = SUM('cleaned_superstore'[Sales])
Total Profit   = SUM('cleaned_superstore'[Profit])
Total Orders   = DISTINCTCOUNT('cleaned_superstore'[Order ID])
Profit Margin  = DIVIDE([Total Profit], [Total Sales], 0)
```

Dashboard Components

- **KPI Cards:** Total Sales, Total Profit, Total Orders, Profit Margin
- **Charts:** Sales by Category, Profit by Region, Monthly Sales Trend, Top 10 Products
- **Slicers:** Year, Region, Segment, Category
- **Dashboard Title:** Superstore Executive Sales Dashboard

Dashboard Preview

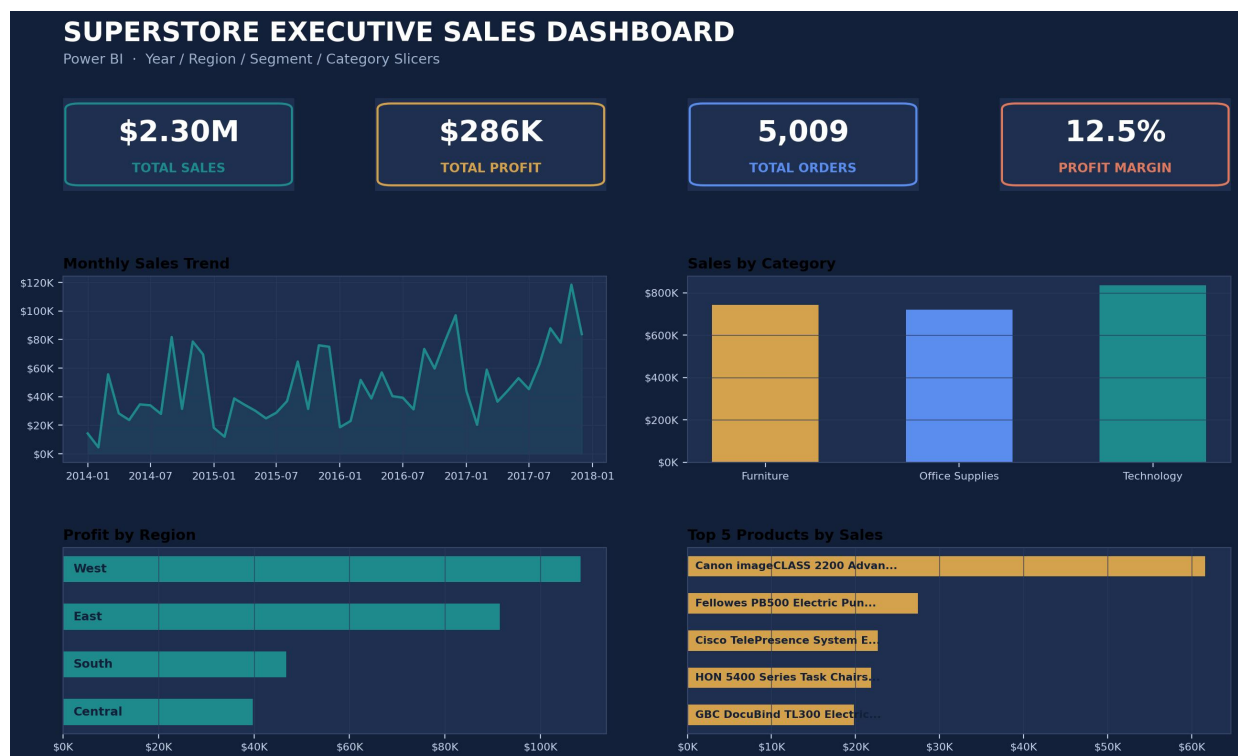


Figure 7. Executive dashboard layout — KPI cards at the top for at-a-glance performance, with trend, category, regional, and product views beneath, all filterable via the Year / Region / Segment / Category slicers.

Key Insights & Recommendations

- **Surface the discount-margin relationship in the dashboard.** Add a Discount slicer or a dedicated 'Discount Impact' page — Figure 4 shows this is one of the single biggest profit levers in the entire dataset and deserves its own view, not just a filter.

- **Drill into Sub-Category, not just Category.** Add a Sub-Category drill-down beneath Category, since Tables and Bookcases are quietly losing money inside a Furniture category that looks fine at the top level.
- **Add a YoY growth KPI.** With 2017 sales up 51% over 2014, add a YoY Growth % measure alongside the existing KPI cards so growth — not just absolute totals — is visible at a glance.

Tech Stack & Skills Summary

Data Cleaning & EDA	Python (Pandas) — missing value & duplicate checks, feature engineering, descriptive statistics
Database & Querying	Microsoft SQL Server — database creation, data import, GROUP BY aggregation, TOP-N and date-part queries
Visualization & BI	Power BI (DAX measures, KPI cards, slicers), Matplotlib/Seaborn for exploratory charts
Business Translation	Insight-to-recommendation storytelling, executive dashboard design, KPI definition

Data source: the classic Superstore retail dataset (9,994 order line items, 2014–2017), cleaned and feature-engineered in Python as described in Phase 2 of this project.

DATA ANALYTICS PROJECT PORTFOLIO

Customer Churn Analysis and Visualization Using Python, SQL Server, and Power BI

An end-to-end analytics workflow — from data cleaning to business-ready dashboards — uncovering the drivers of customer churn and translating them into actionable retention strategy.

PYTHON

PANDAS

SQL SERVER

POWER BI

DAX



Project Objective

The goal of this project was to analyze customer behavior, identify the key factors contributing to customer churn, and translate those insights into practical business recommendations for improving customer retention.

Tools and Technologies Used

TOOL	PURPOSE
Python (Google Colab)	Data cleaning and preprocessing
Pandas	Data manipulation and analysis
SQL Server	Data storage and business analysis queries
Power BI	Dashboard creation and visualization
DAX	KPI calculations and measures

Project Workflow

Phase 1 — Data Collection

- Imported the Customer Churn dataset into Google Colab.
- Examined the dataset structure and columns.

Phase 2 — Data Cleaning (Python)

- Performed missing value checks, data type corrections, duplicate checks, and data transformation.
- Exported the cleaned dataset as **cleaned_customer_churn.csv**.

Phase 3 — SQL Database Development

- Created the **CustomerChurnDB** database and imported **dbo.cleaned_customer_churn**.
- Performed SQL analysis, including:
 - Total Customers — *COUNT(*)*
 - Churn Rate — *GROUP BY Churn*
 - Contract Analysis — *GROUP BY Contract*
 - Payment Method Analysis — *GROUP BY PaymentMethod*
 - High-Risk Customer Segmentation — *WHERE Churn = 1*

Phase 4 — Power BI Dashboard

- Built core KPIs: Total Customers, Churned Customers, Churn Rate, Average Monthly Charges, and Average Tenure.
- Created visualizations for: Churn by Contract, Churn by Internet Service, Churn by Payment Method, Tenure Distribution, and Monthly Charges Analysis.
- Added interactive slicers for Gender, Contract, Internet Service, and Payment Method.

Sample Python Code — Data Cleaning

Below is a representative excerpt of the Python (Pandas) workflow used in Google Colab to clean and prepare the raw churn dataset before loading it into SQL Server.

```
import pandas as pd

# Load raw dataset
df = pd.read_csv("customer_churn.csv")

# 1. Inspect structure
df.info()
df.head()

# 2. Check for missing values
print(df.isnull().sum())
df["TotalCharges"] = df["TotalCharges"].replace(" ", pd.NA)
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"])
df["TotalCharges"].fillna(df["TotalCharges"].median(), inplace=True)

# 3. Fix data types
df["SeniorCitizen"] = df["SeniorCitizen"].astype("category")
df["Churn"] = df["Churn"].map({"Yes": 1, "No": 0})

# 4. Remove duplicate records
df.drop_duplicates(subset="customerID", inplace=True)

# 5. Standardize text fields
text_cols = df.select_dtypes(include="object").columns
df[text_cols] = df[text_cols].apply(lambda x: x.str.strip())

# 6. Export cleaned dataset
df.to_csv("cleaned_customer_churn.csv", index=False)
print(f"Cleaned dataset shape: {df.shape}")
```

The cleaned CSV was then imported into SQL Server as **dbo.cleaned_customer_churn** for business analysis, including churn-rate, contract, and payment-method breakdowns.

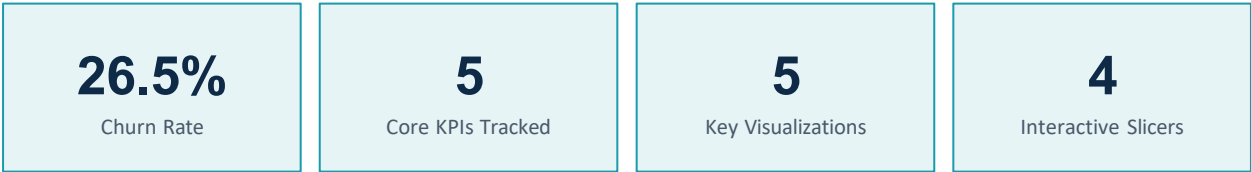
Power BI Dashboard

The final deliverable is an interactive Power BI dashboard summarizing churn KPIs, customer segments, and behavioral patterns, with slicers for on-the-fly filtering.



Customer Churn Analysis Dashboard — Business Intelligence Portfolio Project

Dashboard KPI Snapshot



Key Findings

1. Churn Rate	Approximately 26.5% of customers left the company.
2. Contract Type	Customers on Month-to-Month contracts showed the highest churn rate.
3. Customer Tenure	Customers with shorter tenure were more likely to leave.
4. Payment Method	Certain payment methods had higher churn concentrations.
5. High-Risk Segments	Specific combinations of contract type, internet service, and payment method were identified as high-risk groups.

Business Recommendations

1

Offer discounts and incentives for customers moving from Month-to-Month contracts to longer-term contracts.

2

Implement customer retention campaigns during the first few months of subscription, when churn risk is highest.

3

Develop targeted offers and outreach for high-risk customer segments identified through contract, service, and payment analysis.

4

Monitor customer satisfaction and billing experiences on an ongoing basis to catch early warning signs of churn.

This project demonstrates a complete analytics pipeline — from raw data to cleaned datasets, from SQL-driven business analysis to an interactive Power BI dashboard — resulting in clear, actionable recommendations for reducing customer churn.